Oscar CABRERA ROJAS

+52 55 1483 9468 | oscar.cabrerarojas@gmail.com | linkedin.com/in/oscar-cabrera-1698b0289/ | github.com/Oscar-Cabr

Education

National Autonomous University of Mexico (UNAM) Mexico City, MEX

Bachelor's degree in Computer Engineering *(Expected Graduation) May 2028*

Relevant Courses: *Data Structures and Algorithms I / II,* ***Object Oriented Programming***

Experience

iOS Development Lab Feb. 2026 – Present

Social Service Intern

- Engineered end-to-end iOS applications focused on social impact, identifying community-specific needs and translating them into functional mobile solutions using Swift, SwiftUI, and MVC architecture.

Electrical Technician at Ambar Electricians May 2018 – Present

Senior member of the installation team *Ambar Electricians & Service*

- Installer in residential and commercial electrical installations, as well as the implementation of solar systems.

Swift Changemakers National Hackathon May 2026

National Finalist *Enactus México*

- Ranked **top 3** of 20 local teams, advancing to the *Energy* challenge track at the national finals.

Competitive Achievements

School of Engineering's Competitive Programming Club (CPCFI) June 2024 – Present

General Coordinator*, Academic Team Lead & Competitor* *Engineering School at UNAM*

- Directed a two-phase, two-division programming contest spanning **6 of the 9** National Preparatory School campuses (UNAM), owning scheduling, tiered problem sets, judging logistics and the on-site finals.
- Mentored 50+ competitors by developing tailored pedagogical strategies to simplify complex DSA topics.
- Authored comprehensive lectures for 10+ advanced topics, delivering 30+ technical classes and building 30+ training contests tailored to the students' learning curve and skill progression.

ICPC Subregional Programming Contest Aug. 2024 – Present

International Collegiate Programming Contest

- Committed to rigorous self-directed training in advanced algorithms, data structures and discrete mathematics, mastering complex topics to maintain a competitive edge in national-level contests.
- Collaborated within a high-performance 3-member team to devise and implement optimized C++ solutions under intense pressure, securing a **top 70** rank out of 500+ teams in the Mexico region.

IEEEXtreme 18.0 & 19.0 Programming Competition Oct. 2024 & Oct. 2025

Competitive Programming IEEE Global Challenge

- Competed in the **24-hour** 18th and 19th editions of IEEEXtreme, Mexico section.
- Successfully solved high-difficulty problems hourly, securing a **top 20 rank** among 130 teams in Mexico section.

Projects

Investment in Solar Systems | *Swift, SwiftUI* | Repo *Swift Changemakers National Hackathon* – May 2026

- Architected an intuitive and accessible investment platform, implementing Human Centered AI principles to demystify blockchain and tokenized finance for users with zero cryptocurrency experience through zero-jargon UI and visual transparency of smart contract interactions.
- Integrated on-device generative AI for personalized financial calculations and natural language explanations, hardened transactions through non-custodial embedded wallet key isolation, and prioritized inclusive design.

School Management System | *Java* | Repository October 2024

- Engineered a scalable application, strictly OOP oriented, using concepts as abstraction, inheritance, polymorphism, and encapsulation to create a modular, secure and maintainable architecture.
- Implemented Design Patterns to manage object instantiation and behavior, such as *Singleton* and *Memento*.

Technical Skills

Programming Languages: **C**, **C++**, **Swift**, Java, JavaScript (& React), Python
Developer Tools: Linux (Arch, Debian), Git, GitHub, Shell Scripting, LaTeX, Docker, Neovim, NetBeans
AI Tools: Claude (Code, Design), ChatGPT, Gemini (Gems, Notebook), Opencode
Languages: Spanish (native), English (C1), Mandarin Chinese (HSK1)